INDIAN INSTITUTE OF INFORMATION TECHNOLOGY KOTTAYAM

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STANDARD OPERATING PROCEDURE & USER MANUAL

QForge Platform: System Operating Specification

Comprehensive Technical Manual for Interactive Clinical Web Studio (`app_web.py`) and Laboratory Control Console (`app.py`)

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1.0 INTRODUCTION & ARCHITECTURAL CONTEXT

The **QForge** framework is an end-to-end differentiable Hybrid Quantum-Classical-Quantum Convolutional Neural Network (QCQ-CNN) engineered for binary diagnosis of brain neoplasms from Magnetic Resonance Imaging (MRI). Designed to resolve fundamental limitations in published literature (Long et al., *Nature Scientific Reports*, 2025), QForge introduces:

1. **Adaptive Front-End Quantum Convolution:** Converts frozen, pseudo-random single-qubit rotations into differentiable PyTorch parameters updated via analytical gradients.
2. **Correlated Multi-Observable Pauli Readout:** Augments the terminal 2-qubit Variational Quantum Circuit (VQC) with a joint expectation observable $\langle Z_0 Z_1 \rangle$, preserving two-qubit quantum entanglement correlations prior to decision classification.
3. **NISQ Hardware Noise Modeling:** Integrates depolarizing noise channel simulation ($\gamma \in [0.00, 0.05]$) and finite-shot measurement sampling (128 to 1024 shots) using density matrix formulations.

To facilitate both rigorous laboratory experimentation and intuitive clinical demonstration, the system provides two distinct user interfaces: the **Interactive Web Studio** (`app_web.py`) and the **Desktop Control Workstation** (`app.py`). This document establishes standard operating protocols for both environments.

2.0 SYSTEM PREREQUISITES & ENVIRONMENT SETUP

The platform executes on standard x86-64 host architectures under Python 3.10+ (Python 3.12 verified) using pure-PyTorch vectorized statevector simulation. All quantum operations execute locally without cloud account dependencies.

Subsystem	Specification / Component	Functional Role
Host Runtime	Python 3.12.5 (64-bit)	Core interpreter and virtual environment.
Deep Learning Engine	PyTorch 2.14.1 (CPU execution)	Autograd backpropagation, tensor calculus, vectorized statevector algebra.
Quantum Reference	PennyLane 0.45.1	Analytical quantum node reference, baseline mathematical verification.
Interactive Web Server	Streamlit 1.66.1	Browser runtime for real-time MRI upload, visualization, and noise sweeps.
Desktop GUI Engine	PySide6 6.8.0+	Qt-based desktop window for automated experiment grids and hardware logging.

> **Execution Authority Protocol:** On Microsoft Windows operating systems, ensure PowerShell execution policies permit virtual environment activation by executing: `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass`.

3.0 OPERATING PROTOCOL: INTERACTIVE WEB STUDIO (`APP_WEB.PY`)

The Interactive Web Studio provides a zero-install, browser-based diagnostic cockpit tailored for evaluators, clinicians, and faculty demonstrations.

3.1 Invocation Protocol

Launch the server by executing the following command in PowerShell within the repository root:

```
.\.venv\Scripts\streamlit.exe run app_web.py
```

(Alternatively, execute the automated batch script: `.\run_web.bat`). The runtime will bind to port 8501 and launch your default browser at `http://localhost:8501`.

3.2 Operational Workflows

> **Workflow 1: Acquisition & Spatial Rescaling**
> Navigate to **Tab 1 (`> 01 // LIVE INFERENCE & KERNELS`)**:
>
> - **Standard Presets:** Click `[ LOAD: GLIOMA SAMPLE ]` to load a validated malignant neoplasm scan, or `[ LOAD: MENINGIOMA SAMPLE ]` to load a benign lesion.
> - **Arbitrary Acquisition:** Drag-and-drop any clinical brain MRI scan (PNG/JPG format) into the file uploader.
> - **Deterministic Pipeline:** The system automatically converts inputs to grayscale, resamples to 28×28 via bicubic interpolation, and scales luminance values to the real unit interval $[0.0, 1.0]$.

> **Workflow 2: 4-Channel Quantum Filter Feature Inspection**
> The central display renders the four 14×14 feature map channels produced by the front-end 4-qubit quantum convolutional kernel. Each channel corresponds to the Pauli-Z expectation value on an individual qubit wire:
>
> - `Channel 0 (<Z0>)`: Extracts low-frequency luminance topology.
> - `Channels 1-3 (<Z1>, <Z2>, <Z3>)`: Capture orthogonal, high-frequency boundary contrasts across tumor margins.

> **Workflow 3: Multi-Observable Correlated VQC Evaluation**
> In the left sidebar selector, choose between the three available architectural variants:
>
> - **Model A (Frozen Random Baseline):** Static random filter with terminal single-qubit measurement $\langle Z_0 \rangle$.
> - **Model B (Trainable Filter):** End-to-end trained filter with terminal single-qubit measurement $\langle Z_0 \rangle$.
> - **Model B + Correlated Readout (Full QForge Architecture):** Evaluates the complete joint Pauli vector $\mathbf{M} = [\langle Z_0 \rangle, \langle Z_1 \rangle, \langle Z_0 Z_1 \rangle]^T$. Inspect the displayed joint expectation $\langle Z_0 Z_1 \rangle$ in the lower panel to verify active entanglement correlation preservation.

> **Workflow 4: Real-Time NISQ Noise Stress-Testing**
> In the left sidebar, toggle **Enable Simulated Quantum Noise**:
>
> - Adjust the **Depolarizing Rate γ** slider across the range $[0.00, 0.05]$.
> - Configure the **Measurement Shots** selector ($128, 256, 512, 1024$, or Infinite).

> - Observe diagnostic confidence shifts in real time: Model A exhibits rapid accuracy decay under increasing γ, while the Correlated Architecture demonstrates superior fault tolerance.

4.0 OPERATING PROTOCOL: DESKTOP CONTROL WORKSTATION (`APP.PY`)

The desktop application (engineered in PySide6/Qt) is designed for laboratory operations, multi-seed training scheduling, and telemetry logging.

4.1 Invocation Protocol

```
.\.venv\Scripts\python.exe app.py
```

(Alternatively, double-click `run_app.bat`).

4.2 Subsystem Navigation

Module Tab	Primary Functionality	Key Operator Controls
01. CONTROL	Experiment setup & parameter configuration.	Filter mode selector, learning rate spinboxes, seed specification, `[ RUN TRAINING ]` execution button.
02. EXECUTION	Real-time simulation process monitoring.	Live console output, epoch duration timers, gradient norm tracking, CPU thread saturation gauges.
03. RESULTS	Aggregated performance analytics.	5-seed comparison matrix, paired t-test and Wilcoxon signed-rank tables, CSV export utilities.
04. VISUALIZATIONS	Publication artifact browser.	Dedicated chips for Fig 1 (Dynamics), Fig 2 (Drift), Fig 3 (ROC), Fig 5 (Feature Maps), and **Fig 6 (NISQ Noise)**.

5.0 MATHEMATICAL FORMULATIONS & ACADEMIC DEFENSE GUIDE

This chapter outlines the formal mathematical principles underpinning QForge and provides structured responses to standard technical queries raised during academic evaluations.

5.1 Mathematical Foundations

- **Angle State Preparation:** Classical pixel intensities $p_i \in [0, 1]$ are mapped to 4-qubit states via Pauli-Y rotations: $$|\psi_{\text{enc}}\rangle = \bigotimes_{i=0}^3 R_Y(\pi p_i) |0\rangle^{\otimes 4}$$
- **Differentiable Variational Transformations:** Layered rotations parameterize the transformation: $$U(\mathbf{\theta}, \mathbf{\theta}') = \bigotimes_{i=0}^3 \left[R_Z(\theta'_i) R_Y(\theta_i) \right]$$
- **Joint Observable Correlated Readout:** The joint expectation operator preserves entanglement: $$\langle Z_0 Z_1 \rangle = (P_{00} + P_{11}) - (P_{01} + P_{10})$$
- **Depolarizing Noise Channel:** Gate error is modeled via density matrix degradation: $$\mathcal{E}_{\text{dep}}(\rho) = (1 - \gamma)\rho + \frac{\gamma}{2^n}\mathbf{I}$$

5.2 Technical Defense FAQ (Viva Review)

> **Query 1: Why does QForge report ~87% test accuracy when Long et al. (2025) reported 95.18%?**
> *Defense:* The published baseline utilized raw Kaggle partitions with **110 cross-split duplicates**, causing test-set leakage. Team QForge executed perceptual hash deduplication (`audit_and_deduplicate()`) to eliminate synthetic score inflation. On this verified leak-free dataset, the empirical baseline is 85.75%, which our correlated model elevates to 87.25%.

> **Query 2: How does the Correlated Pauli Readout overcome information bottlenecking?**
> *Defense:* Conventional VQC heads exclusively evaluate single-qubit observable $\langle Z_0 \rangle$, mathematically tracing out wire 1 and discarding non-local quantum correlations synthesized by CNOT gates. Our multi-observable vector $\mathbf{M} = [\langle Z_0 \rangle, \langle Z_1 \rangle, \langle Z_0 Z_1 \rangle]^T$ preserves two-qubit entanglement prior to linear classification.

> **Query 3: What does the NISQ noise simulation demonstrate regarding physical deployability?**
> *Defense:* Under realistic superconducting gate error rates ($\gamma = 0.02 - 0.03$), the frozen-filter baseline degrades by 10.54%, whereas our adaptive trainable architecture retains over 84.5% accuracy (only 2.75% degradation), demonstrating hardware fault resilience.

6.0 QUALITY ASSURANCE & VERIFICATION PROTOCOL

System integrity is continuously verified through an automated `pytest` test harness. To reproduce all verification guarantees locally:

```
.\.venv\Scripts\pytest.exe tests/test_qsim_and_models.py tests/test_noise.py -v
```

Test Routine	Target Module	Verification Criteria	Result
`test_4q_simulator_parity`	`qsim.py` vs PennyLane	Absolute numerical tolerance $\text{atol} < 10^{-5}$	PASSED
`test_correlated_vqc_parity`	`models.py` vs PennyLane	Exact expectation and gradient parity on $[\langle Z_0 \rangle, \langle Z_1 \rangle, \langle Z_0 Z_1 \rangle]$	PASSED

Test Routine	Target Module	Verification Criteria	Result
`test_gradient_flow`	`quantum_filter.py`	Non-zero autograd parameter updates in trainable mode	PASSED
`test_depolarizing_scaling`	`noise.py`	Attenuates expectation values by $(1 - \gamma)$	PASSED
`test_finite_shot_bounds`	`noise.py`	Expectation outputs strictly bounded in $[-1.0, 1.0]$	PASSED

Document Certification: This technical specification conforms to academic project standards for the Department of Computer Science & Engineering and Department of Artificial Intelligence & Data Science at IIIT Kottayam.